

including intercept term is the degrees of freedom. In notations,
 $nR^2 \sim \chi^2_{df}$.

- (v) If the computed χ^2 is less than the critical value at the assumed level of significance, it implies that heteroscedasticity is not present.
- (vi) If the computed χ^2 is greater than the critical value at the assumed level of significance, there is heteroscedasticity in the dataset.

One limitation of the White's Test is that using the squares and cross products of the independent variables as regressors leads to a decrease in the degrees of freedom. Further, the White's test statistic may be found to be statistically significant due to specification error and not necessarily due to heteroscedasticity. Thus, it is a test of both pure heteroscedasticity and/or specification error.

Check Your Progress 2

- 1) What are the various tests for heteroscedasticity?

- 2) Describe how the White's Test can be carried out to test for the presence of heteroscedasticity.

11.5 REMEDIAL MEASURES

The tests discussed above help us in identification of the presence of heteroscedasticity. There are various methods to eliminate heteroscedasticity from a dataset. Some of these methods are discussed below.

1. Weighted Least Squares

There are situations where we have some ideas about the pattern of heteroscedasticity in a dataset. For instance, suppose $\sigma_i^2 = \sigma^2 z_i^2$ and z_i are

known. In this case we can transform the regression equation $y_i = \alpha + \beta x_i + u_i$. We can divide the above regression model throughout by z_i such that

$$\frac{y_i}{z_i} = \alpha \frac{1}{z_i} + \beta \frac{x_i}{z_i} + \frac{u_i}{z_i} \quad \dots (11.21)$$

In this case, let the error term be transformed into $\frac{u_i}{z_i} = v_i$.

We find that the error term v_i has a constant variance σ^2 .

$$Var(v_i) = E\left(\frac{u_i}{z_i}\right)^2 = \frac{\sigma_i^2}{z_i^2} = \sigma^2 \quad \dots (11.22)$$

This method is called the Weighted Least Squares (WLS) method wherein the i^{th} term is weighed by $\frac{1}{z_i}$. Here, the rule is to regress $\frac{y_i}{z_i}$ on $\frac{1}{z_i}$ and $\frac{x_i}{z_i}$ without the constant term.

Let us take another example: $Var(u_i) = \sigma^2 x_i$ and $\alpha = 0$. The equation $y_i = \beta x_i + u_i$ can be transformed by dividing throughout by $\sqrt{x_i}$ such that,

$$\frac{y_i}{\sqrt{x_i}} = \beta \frac{x_i}{\sqrt{x_i}} + \frac{u_i}{\sqrt{x_i}} \quad \dots (11.23)$$

$$\text{In this case, the WLS estimator } \beta^* = \frac{\sum(\sqrt{x_i})(y_i/\sqrt{x_i})}{\sum\sqrt{x_i}^2} = \frac{\bar{y}}{\bar{x}} \quad \dots (11.24)$$

That is, β^* is the ratio of the means. Similarly, if $\sigma_i^2 = \sigma^2 x_i^2$, then we can transform the equation as follows:

$$\frac{y_i}{x_i} = \alpha \frac{1}{x_i} + \beta + v_i \quad \dots (11.25)$$

Here, the constant term β is the slope coefficient of the original equation.

2. Two-Step Weighted Least Squares

If error variances are not known, for instance, $\sigma_i^2 = \sigma^2 [E(y_i)]^2 = \sigma^2 [\alpha + \beta x_i]^2$, a two-stage procedure may be employed. First, using OLS, the estimators $\hat{\alpha}$ and $\hat{\beta}$ are obtained. Second, regress $\frac{y_i}{\hat{\alpha} + \hat{\beta} x_i}$ on $\frac{1}{\hat{\alpha} + \hat{\beta} x_i}$ and $\frac{x_i}{\hat{\alpha} + \hat{\beta} x_i}$ without the constant term. The standard errors obtained for the estimates of α and β are only asymptotically valid because the weights are estimated.

3. Maximum Likelihood Method

In the above example, if the errors are normally distributed, maximise the likelihood function:

$$LogL = -n \log \sigma - \sum \log (\alpha + \beta x_i) - \frac{1}{2\sigma^2} \sum \left(\frac{y_i - \alpha - \beta x_i}{\alpha + \beta x_i} \right)^2 \quad \dots (11.26)$$

Such a method is efficient than the WLS method if the errors are normally distributed. Note that maximising the above likelihood function is not the same as minimizing $\sum \left(\frac{y_i - \alpha - \beta x_i}{\sigma_i} \right)^2$ which is the method of Weighted Least Squares.

4. Generalised Least Squares (GLS)

The OLS procedure gives equal weight to each observation. Ideally, observations with higher variance ought to be given less weight and vice-versa. If σ_i^2 is known, then $y_i = \alpha + \beta x_i + u_i$ can be transformed by dividing throughout by σ_i such that,

$$\frac{y_i}{\sigma_i} = \frac{\alpha}{\sigma_i} + \beta \frac{x_i}{\sigma_i} + \frac{u_i}{\sigma_i}$$

$$\text{Or, } \frac{y_i}{\sigma_i} = \frac{\alpha}{\sigma_i} + \beta \frac{x_i}{\sigma_i} + v_i \quad \dots (11.27)$$

$$\text{where, } v_i = \frac{u_i}{\sigma_i}.$$

$$\text{Var}(v_i) = \text{Var}\left(\frac{u_i}{\sigma_i}\right) = \frac{E(u_i^2)}{\sigma_i^2} = \frac{\sigma_i^2}{\sigma_i^2} = 1 \quad \dots (11.28)$$

Thus, the transformed model has homoscedastic error variance equal to unity. This transformed model satisfies the usual classical assumptions. Applying OLS to this transformed model is known as the Generalised Least Squares (GLS) method and it produces estimators which are best linear unbiased estimator (BLUE). Effectively, in GLS we minimise the weighted residual sum of squares (RSS) given by $\sum w_i \hat{u}_i^2$ where the weight $w_i = \frac{1}{\sigma_i^2}$. This shows that the observations with larger variance get lesser weight and vice-versa. Essentially, this is the Weighted Least Squares procedure which is a special case of the GLS procedure.

The GLS procedure in matrix form is discussed below. Let us take the regression model in matrix form

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}$$

Let us assume that there is no autocorrelation or heteroscedasticity.

In that case,

$$E(\mathbf{u}\mathbf{u}') = \sigma^2 \mathbf{I}.$$

Now let us relax these assumptions, and assume that

$$E(\mathbf{u}\mathbf{u}') = \boldsymbol{\varphi}$$

where $\boldsymbol{\varphi}$ is an arbitrary positive definite matrix. Pre-multiplying $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}$ by $\boldsymbol{\varphi}^{-1/2}$, we get,

$$\mathbf{y}^* = \mathbf{X}^*\boldsymbol{\beta} + \mathbf{u}^* \quad \dots (11.29)$$

$$\text{Here, } \mathbf{y}^* = \mathbf{y}\boldsymbol{\varphi}^{-\frac{1}{2}}$$

$$\mathbf{X}^* = \mathbf{X}\boldsymbol{\varphi}^{-\frac{1}{2}}$$

$$\mathbf{u}^* = \mathbf{u}\boldsymbol{\varphi}^{-\frac{1}{2}}$$

$$\text{Thus, } E(\mathbf{u}^*\mathbf{u}^{*\prime}) = \mathbf{u}\boldsymbol{\varphi}^{-\frac{1}{2}}\mathbf{u}'\boldsymbol{\varphi}^{-\frac{1}{2}} = \mathbf{I} \quad \dots (11.30)$$

Hence, with this method, the error variances become homoscedastic. Such a method is called Generalised Least Squares (GLS). The GLS estimator $\boldsymbol{\beta}_{GLS}$ is given by:

$$\begin{aligned} \boldsymbol{\beta}_{GLS} &= (\mathbf{X}^{*\prime}\mathbf{X}^*)^{-1}(\mathbf{X}^{*\prime}\mathbf{y}^*) = (\mathbf{X}'\boldsymbol{\varphi}^{-\frac{1}{2}}\mathbf{X}\boldsymbol{\varphi}^{-\frac{1}{2}})^{-1}(\mathbf{X}'\boldsymbol{\varphi}^{-\frac{1}{2}}\mathbf{y}\boldsymbol{\varphi}^{-\frac{1}{2}}) \\ &= (\mathbf{X}'\boldsymbol{\varphi}^{-1}\mathbf{X})^{-1}(\mathbf{X}'\boldsymbol{\varphi}^{-1}\mathbf{y}) \quad \dots (11.31) \end{aligned}$$

$\boldsymbol{\beta}_{GLS}$ is the best linear unbiased estimator of $\boldsymbol{\beta}$.

$$\begin{aligned} \text{The variance of } \boldsymbol{\beta}_{GLS} &\text{ is given by } \mathbf{Var}(\boldsymbol{\beta}_{GLS}) = (\mathbf{X}^{*\prime}\mathbf{X}^*)^{-1} \\ &= (\mathbf{X}'\boldsymbol{\varphi}^{-\frac{1}{2}}\mathbf{X}\boldsymbol{\varphi}^{-\frac{1}{2}})^{-1} = (\mathbf{X}'\boldsymbol{\varphi}^{-1}\mathbf{X})^{-1} \quad \dots (11.32) \end{aligned}$$

$\boldsymbol{\varphi}$ is the variance-covariance matrix of the error term $\mathbf{u}$. In the presence of autocorrelation and heteroscedasticity, there are elements in this matrix both on the main diagonal and off it. With heteroscedasticity, the main diagonal elements will reflect unequal variances of the error term. But the structure of this matrix is not known in practice. Hence, estimated GLS procedure (EGLS) is used wherein OLS is employed ignoring the problems of autocorrelation and heteroscedasticity. The residuals $\hat{\mathbf{u}}$ are estimated and they are used to form the estimated variance-covariance matrix of the error term. This matrix is then employed to get the estimated GLS (EGLS) estimator of $\boldsymbol{\beta}$. EGLS estimators are consistent estimators of GLS.

5. White's Estimator

Heteroscedasticity

If the error variances σ_i^2 are not known, the White's Heteroscedasticity corrected standard errors or robust standard errors can be used. In the regression model, since

$$y_i = \beta x_i + u_i, \text{ the variance of } \hat{\beta} \text{ is given by } Var(\hat{\beta}) = \frac{\sum x_i^2 \sigma_i^2}{(\sum x_i^2)^2}.$$

Since σ_i^2 are not known, White suggests using $\hat{u}_i^2$ and estimate,

$$Var(\hat{\beta}) = \frac{\sum x_i^2 \hat{u}_i^2}{(\sum x_i^2)^2} \quad \dots (11.33)$$

This variance is consistent, that is, it converges to $\frac{\sum x_i^2 \sigma_i^2}{(\sum x_i^2)^2}$ with increase in sample size. Hence, this is a large sample procedure. Empirically, White' robust standard errors can be found to exceed or fall short of the OLS standard errors.

White' robust standard errors can be estimated for k-variable regression model as well. Consider, $Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + \dots + \beta_k X_{ki} + u_i$. In this case, the variance of $\hat{\beta}_j$ is given by

$$Var(\hat{\beta}_j) = \frac{\sum \hat{w}_{ji}^2 \hat{u}_i^2}{(\sum \hat{w}_{ji}^2)^2} \quad \dots (11.34)$$

Here, $\hat{w}_j$ is the residual from regressing X_j on remaining regressors (auxillary regression). One limitation of the White's procedure is that the estimators obtained through it may not be as efficient as those obtained by variable transformation methods employed in WLS procedure illustrated above.

6. Logarithmic Transformation

Sometimes, logarithmic transformation of variables can be done to solve the problem of heteroscedasticity. Economic reasoning supports the use of log linear functional form in some cases. For instance, in case of Cobb- Douglas production function $Y = AK^\alpha L^\beta$, the log linear form given by $\log Y = \log A + \alpha \log K + \beta \log L$ is more suitable as it helps estimate the elasticity of output with respect to various inputs. In another example, price and income elasticity can be estimated from logarithmic transformation such as $\log Q = \alpha + \beta_1 \log P + \beta_2 \log Y$ [Q = output, P = price, Y = income].

7. Using Deflators

The variables can be deflated to solve the problem of heteroscedasticity. Sometimes, economic logic supports this procedure, for instance, in problems involving relative prices or real incomes. In other cases, deflation of variables is simply done to solve the problem of heteroscedasticity.

Assume that $Y = \alpha Z + \beta X + u$. Also, it is given that $\sigma_i^2 = \sigma^2 Z^2$.

Dividing $Y = \alpha Z + \beta X + u$ by Z , we get,

$$\frac{Y}{Z} = \alpha + \frac{\beta X}{Z} + \frac{u}{Z}$$

$$\text{Or, } \frac{Y}{Z} = \alpha + \frac{\beta X}{Z} + u' \quad \dots (11.35)$$

Here $u' = \frac{u}{Z}$ and $Var(u') = \sigma^2$

The deflated variables equation $\frac{Y}{Z} = \alpha + \frac{\beta X}{Z} + u'$ should be used to obtain the estimates of α, β and σ^2 and not the original equation. Once the estimators are obtained, the inferences such as residual calculation, future values prediction, etc. should be made on the basis of the following rule. When deflation of variables is supported by economic logic, the deflated variables equation should be used to draw inferences such as residuals calculation, future values prediction, etc. However, the original equation should be used to draw inferences if deflation of variables is simply done to solve the problem of heteroscedasticity.

In case there are many deflated variables equation, one may be tempted to compare different procedures based on the standard errors of coefficients. After all, the purpose of deflation is to arrive at more efficient estimates. But, such a comparison should be avoided because with heteroscedasticity the standard errors of the coefficients are biased. One should be very careful in making such comparisons. Residuals examination will prove more useful.

Such a procedure of deflation of variables may enhance or reduce the correlation between variables but drawing of inferences should not be based on correlations. For instance, in the above example, the correlations between $\frac{Y}{Z}$ and $\frac{X}{Z}$ and that between Y and X should not be compared to decide which equation, the original one or the transformed one is better for making inferences such as residuals

calculation, future values prediction, etc. The presence of a common denominator Z may increase the correlation between $\frac{Y}{Z}$ and $\frac{X}{Z}$, but such a correlation is ‘spurious’ if there is no close relationship between X and Y , a priori. However, the problem of spurious correlation does not exist if the original hypothesis is formulated in ratio terms, for example, relative prices or real output.

Check Your Progress 3

- 1) In the model, $y_i = \alpha + \beta x_i + u_i$, the variance of the error terms depends on a variable z_i . Which of the following three specifications is best suited?

$$\text{Var}(u_i) = \sigma^2$$

$$\text{Var}(u_i) = \sigma^2 x_i$$

$$\text{Var}(u_i) = \sigma^2 x_i^2$$

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11.6 LET US SUM UP

Heteroscedasticity occurs when the variance of the error terms is unequal across observations. In the presence of heteroscedasticity, the OLS estimators are unbiased and consistent but they are no longer efficient. The estimated variances are also biased with heteroscedasticity. Thus, the usual tests of significance are not valid.

Various tests can be applied to detect the presence of heteroscedasticity. The White’s test, Goldfeld-Quandt test and the Breush-Pagan test are widely used. Having detected heteroscedasticity, corrective measures such as Weighted Least Squares (WLS) method can be applied. But, for this, the pattern of heteroscedasticity must be known so that the original equation can be suitably transformed. The White’s Heteroscedasticity Corrected Standard Errors or the robust standard errors are also reported by econometric software; but these can be obtained only for large samples. Sometimes, simple logarithmic transformation of variables can also solve the problem.

11.7 KEY WORDS

Homoscedasticity : The error terms are said to be homoscedastic if the variance of the error term u_i conditional upon x_i (x_i is the independent variable) is equal to σ^2 . $E(u_i^2) = Var(u_i) = \sigma^2 (i = 1, 2, \dots, n)$.

Heteroscedasticity : Error terms have unequal variances. The variance of u_i conditional upon the given x_i is $E(u_i^2) = Var(u_i) = \sigma_i^2$.

Weighted Least Squares : Here, we minimise the weighted Residual Sum of Squares (RSS). Observations with higher variance get less weight and vice-versa.

11.8 SOME USEFUL BOOKS

Brooks, Chris (2014), '*Introductory Econometrics for Finance*,' Third Edition, Cambridge University Press

Gujarati, Damodar N. and Sangeetha (2007), '*Basic Econometrics*,' Fourth Edition, Tata McGraw Publishing Company Limited, New Delhi, Edition 2007

Maddala, G. S. and Kaja Lalihiri (2012), '*Introduction to Econometrics*,' Fourth Edition, John Wiley and Sons Ltd

Verbeek, Marno (2017), '*A Guide to Modern Econometrics*,' Fifth Edition, John Wiley and Sons Ltd.

Wooldridge, Jeffrey (2022), '*Introductory Econometrics*,' Seventh Edition, Cengage Learning

11.9 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Hint: Use Weighted Least Squares technique.
- 2) False. Regression coefficients are unbiased but their standard errors are biased. Refer to Section 11.3, Points 1 and 3.

Check Your Progress 2

- 1) There are a number of tests to test for the presence of heteroscedasticity. Refer to Section 11.4 and answer.
- 2) The steps are given in detail in Section 11.4. Go through it.

- 1) The first specification is best suited because the error terms are homoscedastic in this case. If there is a choice among the remaining two specifications, proceed as follows. First estimate the residuals $\hat{u}_i$. Plot of $\hat{u}_i$ against x_i will give some indication of heteroscedasticity. Apply Glejser's test and regress $\hat{u}_i$ on x_i as follows.

$$|\hat{u}_i| = \alpha + \beta_1 x_i + v_i$$

$$|\hat{u}_i| = \alpha + \beta_2 \sqrt{x_i} + v_i$$

If β 's turn out to be insignificant, then there is no heteroscedasticity. If β_1 is significant, then use $Var(u_i) = \sigma^2 x_i^2$. If β_2 is significant, then use $Var(u_i) = \sigma^2 x_i$. If both β_1 and β_2 are significant, choose the specification, with high R^2 .



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UNIT 12 ERRORS IN VARIABLES*

Structure

- 12.0 Objective
- 12.1 Introduction
- 12.2 Consequences of Errors in Variables
 - 12.2.1 Measurement Errors in Y
 - 12.2.2 Measurement Error in X
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12.0 OBJECTIVES

After going through this Unit you should be in a position to

- explain the concept of errors in variables;
- identify the consequences of errors in variables;
- explain the concept of instrumental variables;
- test for the presence of measurement errors; and
- take remedial measures to solve the problem of measurement error.

12.1 INTRODUCTION

In ordinary least squares model we assume that sample observations are measured accurately. All our formulae are based upon the presumption that variables (both explained and explanatory) are measured without error. The only form of error admitted to our model is in the form of disturbance term. Here the error term represents the influence of various explanatory variables that have not accurately been included in the model. However, the assumption may not be realistic particularly in the case of secondary data.

Variables, both dependent and independent, are measured subject to error. In particular, the available data may not refer to the variable as specified, as specified, as in the case of proxy variable, or there may be systematic biases in

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